



THINGS YOU WANTED TO KNOW!

Ques. WHAT IS HAPTIC TECHNOLOGY? WHAT KIND OF PROJECTS CAN BE MADE USING IT? WHAT TRAINING COURSES ARE OFFERED IN THIS DOMAIN?

Nikhil Lamba

Ans. Haptics is the science of applying touch (tactile) sensation and control to interact with computer applications. By using special input/output devices (joysticks, data gloves and other devices), users can receive feedback from computer applications in the form of sensations felt on a hand or other parts of the body.

In combination with a visual display, haptics technology can be used to train people for tasks requiring hand-eye coordination such as surgery and spaceship maneuvers. It can also be used for games in which you feel as well as see your interactions with images. For example, it could enable you to play tennis with another computer user somewhere else in the world. Both of you can see the moving ball and, by using the haptic device, position and swing your respective tennis rackets and feel the impact of the ball.

You can do many projects using this technology as per application. Some of these are given below:

Medical. Sense of touch and medical simulation may prove especially useful for training.

Military. For certain applications, for example, where terrain or texture information needs to be conveyed, haptics may be the most efficient for communication.

Assistive technology for the visually impaired. A visually-disabled person can feel things like maps that are displayed on the Internet and also learn computing work by tracing touchable mathematical courses.

Museum displays. This technology

provides the possibility of widening access to information and artefacts held in museums by allowing visitors who live far away to feel objects at a distance, and to show the range of artefacts that are currently in storage due to lack of space.

Entertainment. It can also be used to enhance the gaming experience.

Haptic based trainings and workshops assist engineers to learn about this technology in detail. You may refer the following links for course structure, schedule and topics covered during these trainings.

www.skyfilabs.com/workshops/haptic-arm-arduino-workshop

www.robokart.com/index.php?route=product/product&product_id=391

<https://lagunita.stanford.edu/courses/SelfPaced/Haptics/2014/about>
www.i3indyatechnologies.com/workshop/robotics-arm-development.html

There are no prerequisites for a haptic training, but a basic knowledge of information technology is desirable.

Q2. WHICH SINGLE-BOARD COMPUTER AND MICROCONTROLLER IS IDEAL FOR REAL-TIME AUDIO-VIDEO STREAMING/SURVEILLANCE?

Prashant

A2. Audio-video streaming/surveillance systems play an increasingly important role for maintaining social security. With the development of information technology, especially in embedded CPUs of open Linux systems, video conferencing, remote monitoring and mass data-processing fields can apply embedded technology.

Single-board computer. The hardware system includes a processor, video-capture devices, audio jack for audio and a router to receive video in-

formation through Wi-Fi. You can use any single-board microcomputer like Raspberry Pi B+, Samsung S3C6410 or any other ARM11 processors for this.

Microcontroller. If you only need a basic video feed, consider using motion JPEG to send data across the network. Interface your camera with the microcontroller and run a Web server to send a series of JPEG images to a browser. The quality and frame-rate are not so great, but it lights on processing if the camera outputs JPEGs. You may refer to the article published in *Electronics For You* using the link given below:

www.electronicsforu.com/electronicsforu/circuitarchives/view_article.asp?sno=1505&article_type=1&id=924&tt=unhot

This project is built around Wi-Fi module MRF24WB0MA, PIC microcontroller PIC18F4620 and a camera module using TCP-IP protocol. The camera captures live video from a remote location, which is transmitted wirelessly through the Internet and can be viewed on a PC or laptop.

Q3. WHAT ARE THE USES AND DIFFERENCES BETWEEN IC ULN2003 AND IC ULN2803?

Mahesh J. Shetty

A3. Instead of using individual darlington transistors, you can use an integrated circuit (IC) specially designed for driving high-current loads from TTL-level inputs. Most common ICs of this type are ULN2003, which has seven darlington drivers in a 16-pin DIP package, and ULN2803, which has eight drivers in an 18-pin DIP package.

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